

Week 7 Lab: Intruders, Password Attacks, and Defensive Monitoring

Connect this week's lecture on intruders with practical defensive security labs in TryHackMe. You will explore how attackers gain access through password attacks and phishing, and how defenders use traffic analysis and intrusion detection to identify suspicious activity. All tasks should be completed inside TryHackMe (AttackBox or your own VM connected to THM).

Learning objectives

The aim of this lab is to help students understand the threat posed by intruders and the defensive controls used to detect or reduce that risk. By completing the activities, students will classify common intrusion methods, including phishing, password guessing, and insider misuse; recognise how packet capture and traffic analysis reveal suspicious behaviour; explain the role of IDS and IPS technologies; and relate password hygiene and multi-factor authentication to real-world defensive practice. Finally, students will connect user-focused attack vectors with network-focused detection methods and reflect on suitable countermeasures for each scenario.

Requirements

- TryHackMe account and access to AttackBox or your own VM.
- Comfort with a terminal and basic packet inspection.
- A short set of written notes for each activity (screenshots optional).

TryHackMe links

- **Phishing Analysis:** <https://tryhackme.com/module/phishing>
- **IDS Fundamentals:** <https://tryhackme.com/room/idsfundamentals>
- **Snort:** <https://tryhackme.com/room/snort>
- **Wireshark: The Basics:** <https://tryhackme.com/room/wiresharkthebasics>
- **Wireshark: Traffic Analysis:** <https://tryhackme.com/room/wiresharktrafficanalysis>
- **Traffic Analysis Essentials (optional refresher):**
<https://tryhackme.com/room/trafficanalysissentials>

Suggested student workflow

1. Start with Phishing Analysis to identify social-engineering indicators and common email red flags.
2. Use Wireshark: The Basics and Traffic Analysis Essentials to refresh packet inspection and protocol awareness.
3. Complete Wireshark: Traffic Analysis to investigate suspicious patterns such as scans, poisoning, or cleartext protocols.
4. Finish with IDS Fundamentals and Snort to connect observed network behaviour with rule-based detection and prevention.

Note: Some rooms may require a TryHackMe Premium subscription.